

RAG-Based Information Retrieval System

Approach & Methodology Report

1. Objective

The objective of this project was to develop a Retrieval-Augmented Generation (RAG) based information retrieval system capable of answering natural-language questions from a PDF document.

The system is designed to retrieve and answer questions across three types of information:

1. Text
2. Tables
3. Figures

The system follows a modular pipeline consisting of document ingestion, content processing, vector and keyword indexing, hybrid retrieval, generation, and a user interface. Its quality was measured with an automated test suite and a question benchmark, reported in Section 10. I used 10 SEC filing documents for development while testing metrics are from the holdout Apple's Q3 2022 Form 10-Q.

2. Approach & Methodology

The system follows the following pipeline:

PDF → Content Extraction → Chunking & Processing → Embedding + BM25 Index → Hybrid Retrieval (Dense + BM25) → LLM → Answer + Sources

The implementation is divided into independent modules so that components such as the embedding model, vector store, or LLM can be replaced without redesigning the entire system.

- **Document Ingestion:** PyMuPDF extracts text/images, pdfplumber extracts tables, and a vision model describes figures, with page and content metadata preserved.
- **Text & Table Representation:** Text is chunked into 800-character sections with overlap, while tables are split into compact row groups with page, title, and header context, keeping each chunk meaningful and within the 256-token embedding limit.
- **Figure Processing:** Embedded images above 40 px are passed to a vision model for concise descriptions, which are stored as searchable chunks with page and image metadata, enabling figure-based retrieval alongside text and tables.

3. Key Design Decisions

- **Unified Retrieval Index:** Text, tables, and figure descriptions share one index, allowing relevant content types to be retrieved without query classification.
- **Local Embeddings:** all-MiniLM-L6-v2 avoids API costs, rate limits, and unnecessary external dependencies.
- **Context-Aware Table Chunks:** Tables are split into row-level chunks with titles and headers, keeping each chunk interpretable and within the embedding limit.
- **Figure Captioning:** Images are captioned during ingestion, making them searchable while reducing query-time processing.

- **Lightweight Vector Store:** A NumPy-based store is sufficient for the small single-document dataset and avoids unnecessary infrastructure.
- **Modular Architecture:** Ingestion, indexing, retrieval, generation, and UI are separated for easier maintenance and component replacement.
- **Hybrid Retrieval:** Dense search handles semantic similarity, while BM25 improves retrieval of exact figures, labels, and rare terms.

4. Assumptions

- The reference document is predominantly text and table based. No reason to engage OCR compute if there is no requirement of it.
- The primary use case is document-based question answering over financial filings.
- Free-tier LLM availability and model names may change, so model configuration is kept separate from the core application logic.

5. Challenges

Financial Table Extraction

Financial statements can contain dense, borderless tables that are challenging for automated PDF table extraction.

The extraction strategy was tested against the actual financial statements, and the default pdfplumber strategy was selected because it preserved the underlying cell values more reliably for this document.

Extraction proved not to be the main weakness: pdfplumber and PyMuPDF's table finder both detect the same 31 tables in the reference document. The measured problem was how tables were chunked and embedded (oversized chunks and missing headers), which was addressed by the row-level chunking

Model/API Behavior

The selected financial language model uses a reasoning process that can consume part of the completion-token budget.

During testing, an insufficient token limit resulted in an incomplete response. This was addressed by increasing the completion budget and adding defensive handling for null responses.

In the baseline evaluation, two answers failed this way; in both cases retrieval had not surfaced the relevant chunk, so the model reasoned at length without an answer to find. No such failures occurred after retrieval was improved.

Numeric Questions and Dense Retrieval

With dense-only retrieval, four of the fourteen development questions were never retrieved, most of them requests for a specific figure. Embedding similarity does not reliably separate two chunks that differ mainly in their numbers. Hybrid retrieval and better table chunks addressed this (Section 10).

Free-Tier API Limits

The free OpenRouter tier allows about 50 requests per day. The live answer-accuracy test is therefore limited to 26 questions, and treats an HTTP 429 response as "not evaluated" rather than as a wrong answer. Repeated live runs during development exhausted one day's quota. Hence the 46 test questions.

6. Evaluation & Results

The system was evaluated on the reference document (Apple Q3 2022 Form 10-Q, 28 pages) with an automated test suite and a benchmark of known questions.

6.1 Method

- **Test suite.** 160 automated tests cover chunking, table conversion, the vector store, BM25 and hybrid search, PDF ingestion on the real document, and the answer prompt.
- **Question benchmark.** 46 questions with evidence strings copied verbatim from the PDF.
- **Relevance.** A retrieved chunk is relevant if it contains every evidence string of the question, so relevance does not depend on the page or chunk type.
- **Metrics.** hit@k (a relevant chunk appears in the top k), precision@5 and mean reciprocal rank (MRR) for retrieval; for answers, whether the generated answer contains the expected value and cites a page.

6.2 Retrieval Results

Table 1. Development questions: dense-only baseline versus the current system (top 5)

Metric	Dense-only baseline	Current system
hit@1	0.29	0.50
hit@3	0.57	0.71
hit@5	0.71	1.00
MRR	0.43	0.66

Table 2. Current system by question split (top 5)

Split	Questions	hit@1	hit@3	hit@5	P@5	MRR
Development	14	0.50	0.71	1.00	0.34	0.66
Held-out	12	0.50	0.67	0.83	0.30	0.61
Extra	20	0.30	0.75	0.90	0.28	0.52
All	46	0.41	0.72	0.91	0.30	0.59

The development score is optimistic because the settings were tuned on those questions. The held-out and extra splits are the more honest measures, and they agree with each other (hit@5 of 0.83 and 0.90), which indicates that the gains generalise. The baseline was only measured on the development questions, so the improvement on unseen questions is inferred rather than compared like for like.

6.3 Answer Accuracy

Answers were generated live with inclusionai/ling-3.0-flash-fin (free tier) and scored by checking that the expected value appears in the answer and that a page is cited.

Table 3. Live answer accuracy

Run	Correct	Cites a page	Model errors
Baseline (development, 14 questions)	8/14 (0.57)	0.71	4 (2 not in excerpts, 2 token budget)
After retrieval fixes (26 questions)	24/26 (0.92)	1.00	0

The two answers marked wrong after the fixes are limitations of the strict string-match scorer, not wrong answers: one gave "\$11.0 billion" for a figure the filing prints as 10,982 million, the other said the effective tax rate "increased" where the expected word was "higher". True accuracy is therefore likely higher than 0.92. Each condition was a single run at temperature 1.0 on a free-tier model, so some run-to-run variation is

expected. Grading on this in an automated way would involve more token usage to prove the accuracy of the model which just not worth it in my opinion.

6.4 Progress and What Drove It

The baseline diagnosis found that 11 of 31 table chunks exceeded the embedding window, that table chunks lacked titles and column headers, and that dense-only search blurs exact figures. Two changes addressed this: row-level table chunks with context, and hybrid dense + BM25 retrieval. A small parameter sweep chose the final settings:

Table 4. Tuning sweep (hit@5)

Table chunk size	Context size	Candidates	Development	Held-out
350 chars	120 chars	20	0.86	0.75
350 chars	300 chars	20	0.93	0.67
200 chars	300 chars	20	0.93	0.75
200 chars (chosen)	300 chars	50	1.00	0.83

Two hypotheses were wrong and were corrected by measurement. First, table extraction was suspected as the cause, but both libraries find the same 31 tables; the problem was chunking and embedding. Second, removing the raw text that duplicates each table looked like it would reduce noise, but it lowered recall (best overall hit@5 0.85 versus 0.92), so it was reverted. One automated test also caught an oversized first setting, because numbers tokenise into many tokens.

6.5 Test Authorship and Verification

Test cases were developed with assistance from Claude Haiku subagents, which were used to generate and extend the test suite, including additional development and held-out questions. Haiku wrote 160 tests and 46 benchmark questions. Their reports were not taken at face value; the results were independently re-checked.

7. Improvements & Future Work

- The current implementation does not contain OCR and relies on the embedded text in the pdf. This should be the first thing to be added to increase broader support.
- It processes one document at a time. A future version could maintain separate indexes for multiple documents and support cross-document retrieval.
- A larger embedding model with a longer input window than the current 256 tokens could improve ranking of numeric and table content.
- Charts drawn as vector graphics are not detected as images. Detecting clusters of drawing operations and rendering those page regions would allow them to be captioned by the vision model.
- For a larger document collection, the NumPy-based store could be replaced with FAISS, Chroma, or another production-oriented vector database. There really was no reason to involve it here due to the scale

12. Conclusion

The main focus of the implementation was to keep the architecture modular and lightweight while maintaining grounded answers with source attribution.

PDF extraction → structured content processing → local embeddings + BM25 → hybrid retrieval → LLM-based generation → source attribution

This provides a practical foundation for document-based information retrieval and can be extended to support larger document collections, improved table understanding and reranking.

On the reference document, retrieval placed a relevant chunk in the top five for 83–90% of unseen questions (hit@5), and live answers were correct and page-cited in 24 of 26 cases. The main gains came from measuring first: row-level table chunks and hybrid retrieval each addressed a weakness identified by the benchmark.

13. Screenshots

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whether the Registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company, or an emerging growth company.
See the definitions of "large accelerated filer," "accelerated filer," "smaller reporting company," and "emerging growth company" in Rule 12b-2 of the Exchange Act.

The Registrant is a large accelerated filer.
Per the cover page of the filing (page 1), the checkbox next to "Large accelerated filer" is marked with ☒, while all other categories — Accelerated filer, Non-accelerated filer, Smaller reporting company, and Emerging growth company — are left unchecked (☐).

- Large accelerated filer: ☒ (page 1)
- Accelerated filer: ☐ (page 1)
- Non-accelerated filer: ☐ (page 1)
- Smaller reporting company: ☐ (page 1)

text - page 1 (score 0.74)

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text - page 26 (score 0.60)

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gross margin and gross margin percentage for the three- and nine-month periods

Gross Margin and Gross Margin Percentage (dollars in millions)

Period	Segment	Gross Margin	Gross Margin %
Three months ended June 25, 2022	Products	\$21,870	34.5%
	Services	\$14,015	71.5%
	Total	\$35,885	43.3%
Three months ended June 26, 2021	Products	\$23,049	36.0%
	Services	\$12,206	69.8%
	Total	\$35,255 ↓	43.3%